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Benchmarking Numerical Methods for Electron Repulsion Integrals: From Direct Summation to Poisson–NUFFT Solvers

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Abstract

Electron repulsion integrals (ERIs) constitute the primary computational bottleneck in quantum chemistry simulations, scaling as $\mathcal{O}(N^4)$ in conventional analytical approaches. This report presents a systematic benchmark of seven numerical methods for computing ERIs, ranging from brute-force direct summation in real space through uniform Fast Fourier Transforms (FFT) to Non-Uniform Fast Fourier Transform (NUFFT) techniques and a two-step Poisson–NUFFT solver. We evaluate each method on the hydrogen molecule H_2 using the `def2-svp` basis set, measuring absolute error against the analytical PySCF reference, wall-clock time, and peak memory consumption as the grid resolution parameter N increases. Our results demonstrate that the Poisson two-step method with a spherical k-space grid achieves the fastest convergence, reaching errors below 10^{-3} at moderate grid sizes, while maintaining competitive computational cost. We discuss the mathematical foundations of each approach, analyse convergence behaviour, and suggest directions for future work including extension to multi-centre integrals and GPU acceleration.

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1 Introduction

Many advances in materials science, battery technology, semiconductor design, and superconductivity owe their progress to quantum chemistry simulations. At their core, these simulations seek approximate solutions to the many-body Schrödinger equation:

$$\hat{H}\,\Psi(\mathbf{r}_1,\ldots,\mathbf{r}_n) = E\,\Psi(\mathbf{r}_1,\ldots,\mathbf{r}_n),$$

where $\hat{H}$ is the molecular Hamiltonian and Ψ is the many-electron wavefunction. Under the Born–Oppenheimer approximation, the nuclei are treated as stationary point charges and the electronic Hamiltonian becomes:

$$\hat{H}_{\text{elec}} = -\frac{1}{2}\sum_i \nabla_i^2 - \sum_{i,A}\frac{Z_A}{|\mathbf{r}_i-\mathbf{R}_A|} + \sum_{i<j}\frac{1}{|\mathbf{r}_i-\mathbf{r}_j|},$$

where the three terms represent the kinetic energy, nuclear–electron attraction, and electron–electron repulsion respectively. The final term is the most computationally demanding.

To discretise the problem, the molecular orbitals are expanded in a basis of K atom-centred functions $\{\chi_\mu\}$:

$$\phi_i(\mathbf{r}) = \sum_{\mu=1}^{K} C_{\mu i}\,\chi_\mu(\mathbf{r}).$$

Gaussian-Type Orbitals (GTOs) are the standard choice because the product of two Gaussians centred at different points yields another Gaussian (the Gaussian Product Theorem), which simplifies integral evaluation:

$$\chi_\mu(\mathbf{r}) = N_\mu\,(x-A_x)^{l_x}(y-A_y)^{l_y}(z-A_z)^{l_z}\,\exp\!\big(-\alpha_\mu|\mathbf{r}-\mathbf{A}|^2\big).$$

Substituting section 1 into the energy expression produces four classes of molecular integrals. The overlap integrals $S_{\mu\nu}$, kinetic energy integrals $T_{\mu\nu}$, nuclear attraction integrals $V_{\mu\nu}$, and the four-centre electron repulsion integrals (ERIs):

$$(\mu\nu|\lambda\sigma) = \iint \frac{\chi_\mu(\mathbf{r}_1)\chi_\nu(\mathbf{r}_1)\;\chi_\lambda(\mathbf{r}_2)\chi_\sigma(\mathbf{r}_2)}{|\mathbf{r}_1-\mathbf{r}_2|}\,d^3r_1\,d^3r_2.$$

For a basis of size K, the number of unique ERIs scales as $\mathcal{O}(K^4)$, making their evaluation the dominant computational bottleneck.

A key insight is that the Coulomb kernel has a simple form in Fourier (reciprocal) space. The Fourier transform of $1/|\mathbf{r}|$ in three dimensions is:

$$\mathcal{F}\left[\frac{1}{|\mathbf{r}|}\right](\mathbf{k}) = \frac{4\pi}{|\mathbf{k}|^2}.$$

This transforms the six-dimensional integral in section 1 into a product of three-dimensional Fourier transforms, reducing it by the convolution theorem to:

$$(\mu\nu|\lambda\sigma) = \frac{1}{(2\pi)^3}\int \frac{4\pi}{|\mathbf{k}|^2}\;\widetilde{f}(\mathbf{k})\;\widetilde{g}^*(\mathbf{k})\,d^3k,$$

where $\widetilde{f}(\mathbf{k})$ and $\widetilde{g}(\mathbf{k})$ are the Fourier transforms of the overlap densities $f(\mathbf{r}) = \chi_\mu(\mathbf{r})\chi_\nu(\mathbf{r})$ and $g(\mathbf{r}) = \chi_\lambda(\mathbf{r})\chi_\sigma(\mathbf{r})$.

The Coulomb potential generated by a charge distribution $\rho(\mathbf{r})$ satisfies Poisson's equation:

$$\nabla^2 V(\mathbf{r}_1) = -4\pi\,\rho(\mathbf{r}_1),$$

with the solution:

$$V(\mathbf{r}_1) = \int \frac{\rho(\mathbf{r}_2)}{|\mathbf{r}_1 - \mathbf{r}_2|}\, d^3 r_2.$$

Fourier-transforming section 1 converts the differential equation into the algebraic relation:

$$\widetilde{V}(\mathbf{k}) = \frac{4\pi}{|\mathbf{k}|^2}\,\widetilde{\rho}(\mathbf{k}),$$

which replaces a costly real-space integration with a pointwise multiplication in k-space. This is the foundation of the two-step Poisson–NUFFT solver benchmarked in this study.

The standard FFT operates on uniformly spaced grids and achieves $\mathcal{O}(N \log N)$ complexity. However, molecular systems are better described by non-uniform grids that concentrate points near the nuclei. The Non-Uniform Fast Fourier Transform (NUFFT) generalises the FFT to handle arbitrary point distributions. It computes sums of the form:

$$F(\mathbf{k}_j) = \sum_{m=1}^{M} c_m\, e^{-i\,\mathbf{k}_j \cdot \mathbf{x}_m},$$

where $\{\mathbf{x}_m\}$ are non-uniform source points and $\{\mathbf{k}_j\}$ are non-uniform target points (this is the type-3 NUFFT). The algorithm achieves $\mathcal{O}(M \log M + N \log N)$ complexity via interpolation onto an auxiliary uniform grid, FFT, and deconvolution [**?**].

1.1 Objectives

The aim of this study is to benchmark seven numerical approaches for computing ERIs, categorised as follows:

1. **Direct summation** on Cartesian and Becke grids in real space,
2. **Uniform FFT** Poisson solver on a Cartesian grid,
3. **One-step NUFFT** with Cartesian and spherical k-space grids,
4. **Two-step Poisson–NUFFT** with Cartesian and spherical k-space grids.

We evaluate convergence behaviour, computational cost, and memory usage, and discuss why certain methods outperform others from a mathematical perspective.

2 Theoretical Background

To evaluate integrals numerically, we replace the continuous integral with a weighted sum over grid points:

$$\int f(\mathbf{r})\, d^3 r \approx \sum_{i=1}^{N} w_i\, f(\mathbf{r}_i),$$

where the choice of points $\{\mathbf{r}_i\}$ and weights $\{w_i\}$ determines both accuracy and efficiency.

The simplest approach tiles a cubic domain $[-L, L]^3$ with N^3 equally spaced points. The spacing is $\Delta x = 2L/(N-1)$ and each weight is $w_i = (\Delta x)^3$. While straightforward, Cartesian

grids waste points in empty regions far from the molecule and lack resolution near nuclei where the density varies rapidly. The error converges polynomially with N.

Becke's partitioning scheme [?] decomposes the molecular integral into atomic contributions:

$$\int f(\mathbf{r})\, d^3r = \sum_A \int w_A(\mathbf{r})\, f(\mathbf{r})\, d^3r,$$

where $w_A(\mathbf{r})$ is the Becke weight function centred on atom A, satisfying $\sum_A w_A(\mathbf{r}) = 1$. Each atomic integral is evaluated on a product grid of radial (Gauss–Legendre or Gauss–Chebyshev) and angular (Lebedev) quadrature points, mapped via a radial transform $r = R\,h(x)$ that concentrates points near the nucleus. The Becke radial transform used in this work maps:

$$r = R\,\frac{1+x}{1-x}, \quad x \in [-1, 1],$$

where R is a scaling parameter (set to 1.5 bohr in our implementation). This naturally places more grid points in the chemically important region near the nuclei.

In reciprocal space, we construct a spherical grid using a product of Gauss–Legendre radial quadrature on $[0, k_{\max}]$ and Lebedev angular quadrature. The volume element in spherical coordinates is $d^3k = k^2\, dk\, d\Omega$, so the quadrature weights include the Jacobian factor k^2:

$$w_j = w_j^{\text{rad}}\, w_j^{\text{ang}}\, k_j^2.$$

A key advantage of this grid for Coulomb integrals is that the k^2 factor in the weights cancels analytically with the $1/k^2$ in the Coulomb kernel (section 1), eliminating the singularity at $k = 0$. Moreover, because Gauss–Legendre nodes on $(0, k_{\max}]$ never include $k = 0$ exactly, the computation is numerically stable without any special treatment.

2.1 Method Formulations

The brute-force approach discretises section 1 directly:

$$(\mu\nu|\lambda\sigma) \approx \sum_{i=1}^{N} \sum_{j\neq i}^{N} \frac{\rho_\mu\nu(\mathbf{r}_i)\, w_i\; \rho_{\lambda\sigma}(\mathbf{r}_j)\, w_j}{|\mathbf{r}_i - \mathbf{r}_j|},$$

where $\rho_{\mu\nu}(\mathbf{r}) = \chi_\mu(\mathbf{r})\chi_\nu(\mathbf{r})$ is the overlap density. The self-interaction term ($i = j$) is excluded by setting the diagonal of the distance matrix to infinity. This method has $\mathcal{O}(N^2)$ complexity, where N is the number of grid points, making it prohibitively expensive for large grids. Two variants are tested: one using a uniform Cartesian grid and one using the Becke molecular grid.

For a uniform Cartesian grid with spacing Δx, the density is Fourier-transformed via the standard 3D FFT:

$$\widetilde{\rho}(\mathbf{k}) = (\Delta x)^3 \sum_{\mathbf{n}} \rho(\mathbf{r_n})\, e^{-i\mathbf{k}\cdot\mathbf{r_n}}.$$

The ERI is then computed via Parseval's theorem applied to the Coulomb-weighted power spectrum:

$$\text{ERI} = \frac{(\Delta k)^3}{(2\pi)^3} \sum_{\mathbf{k}} \frac{4\pi}{|\mathbf{k}|^2}\, |\widetilde{\rho}(\mathbf{k})|^2,$$

where $\Delta k = 2\pi/(N\Delta x)$. The zero-frequency component ($\mathbf{k} = \mathbf{0}$) is excluded to avoid the Coulomb singularity. The FFT achieves $\mathcal{O}(N^3 \log N)$ complexity for an N^3 grid. However, it requires a uniform grid, preventing adaptive refinement near nuclei.

The one-step NUFFT method uses a Becke grid in real space and transforms directly to either a Cartesian or spherical k-space grid via the type-3 NUFFT:

$$\widetilde{\rho}(\mathbf{k}_j) = \mathrm{NUFFT}_3[\rho(\mathbf{r}_i)\, w_i;\ \mathbf{r}_i \to \mathbf{k}_j]\,.$$

The ERI is then:

$$\mathrm{ERI} = \frac{1}{(2\pi)^3} \sum_j w_j^{(k)} \frac{4\pi}{|\mathbf{k}_j|^2} |\widetilde{\rho}(\mathbf{k}_j)|^2.$$

For the spherical k-grid variant, the cancellation $w_j^{(k)} \propto k_j^2$ with $1/k_j^2$ simplifies the expression and removes the singularity, as discussed in section 2.

This method implements the Poisson equation approach from [?]. Given two overlap densities $f(\mathbf{r})$ and $g(\mathbf{r})$, the algorithm proceeds in four steps:

Algorithm 1 Two-Step Poisson–NUFFT

1: Evaluate $g(\mathbf{r}_i)$ on the real-space Becke grid.
2: **Forward NUFFT:** $\widetilde{G}(\mathbf{k}_j) = \mathrm{NUFFT}_3\big[g(\mathbf{r}_i)\, w_i;\ \mathbf{r}_i \to \mathbf{k}_j\big]$.
3: **Apply Coulomb kernel:** $\widetilde{V}^{(g)}(\mathbf{k}_j) = \frac{4\pi}{|\mathbf{k}_j|^2}\, \widetilde{G}(\mathbf{k}_j)$.
4: **Inverse NUFFT:** $V^{(g)}(\mathbf{r}_i) = \frac{1}{(2\pi)^3}\, \mathrm{NUFFT}_3\big[\widetilde{V}^{(g)}(\mathbf{k}_j)\, w_j^{(k)};\ \mathbf{k}_j \to \mathbf{r}_i\big]$.
5: **Numerical integration:** $\mathrm{ERI} = \sum_i f(\mathbf{r}_i)\, V^{(g)}(\mathbf{r}_i)\, w_i$.

The advantage of this formulation is twofold. First, it decouples the two density distributions f and g, which is essential when $f \neq g$ (i.e., for general four-centre integrals). Second, the intermediate potential $V^{(g)}(\mathbf{r})$ is a smooth function in real space, making the final numerical integration highly accurate.

3 Computational Methodology

3.1 Software Environment

All computations were performed in Python 3.13 using the following libraries:

- **PySCF** [?]: Analytical ERI computation via `mol.intor('int2e')` as the reference baseline.
- **NumPy/SciPy**: Array operations, standard FFT, and Gauss–Legendre quadrature.
- **FINUFFT** [?]: Flatiron Institute Non-Uniform FFT library for type-3 transforms.
- **qc-grids**: Construction of Becke molecular grids and Lebedev angular grids.

3.2 Test System

The benchmark molecule is H_2 with a bond length of 0.74 Å (1.40 bohr), using the `def2-svp` basis set [?]. This basis set was selected for its balance of efficiency and accuracy, and its built-in effective core potential (ECP) support for heavier elements. We compute the

simplest ERI, (00|00), where index 0 corresponds to the first GTO basis function. The analytical reference value computed by PySCF is used as the ground truth.

3.3 Density Evaluation

The overlap density for the (00|00) integral reduces to $\rho(\mathbf{r}) = |\phi_0(\mathbf{r})|^2$, where ϕ_0 is the first basis function. This is evaluated using PySCF's `eval_gto` function:

```python
def evaluate_density(mol, points):
    """Evaluate |phi_0(r)|^2 at a set of real-space points."""
    ao = mol.eval_gto("GTOval", points)
    return ao[:, 0] * ao[:, 0]
```

Listing 1: Density evaluation on arbitrary grid points.

3.4 Grid Construction

Three types of grids are employed, summarised in table 1.

Table 1: Grid types used in the benchmark.

Grid	**Space**	**Type**
Uniform Cartesian	Real / k	Equally spaced cube $[-L, L]^3$
Becke (molecular)	Real	Atom-centred radial $\times$ angular
Spherical	k	Gauss–Legendre radial $\times$ Lebedev angular

The **Cartesian grid** spans $[-L, L]^3$ with $L = 12$ bohr for real-space methods and $L = 30$ bohr for k-space grids, using N or $2N$ points per dimension. The larger k-space box ensures adequate frequency resolution.

The **Becke grid** is constructed using the `qc-grids` library with N radial Gauss–Legendre points per atom, a Becke radial transform with $R = 1.5$ bohr, and the "fine" angular preset. Points beyond 12 bohr from the origin are clipped to reduce memory usage without sacrificing accuracy, since the GTO density decays exponentially.

The **spherical k-grid** uses N Gauss–Legendre radial points on $[0, k_{\max}]$ with $k_{\max} = 30\ \text{bohr}^{-1}$, and Lebedev angular grids with degree $\ell = \min(\max(2N, 15), 131)$. The angular degree is decoupled from N and capped at 131, the maximum available Lebedev grid order.

The construction of the spherical k-space grid involves mapping Gauss–Legendre nodes from $[-1, 1]$ to $[0, k_{\max}]$:

$$k_i = \frac{k_{\max}}{2}(x_i + 1), \qquad w_i^{(\text{rad})} = w_i^{(\text{GL})} \frac{k_{\max}}{2},$$

where $(x_i, w_i^{(\text{GL})})$ are the standard Gauss–Legendre nodes and weights. The full three-dimensional grid is formed as the outer product of radial and angular grids:

$$\mathbf{k}_{ij} = k_i\, \hat{\mathbf{n}}_j, \qquad w_{ij} = w_i^{(\text{rad})}\, w_j^{(\text{ang})}\, k_i^2,$$

where $\hat{\mathbf{n}}_j$ and $w_j^{(\text{ang})}$ are Lebedev unit vectors and weights.

```python
def generate_spherical_k_grid(n_rad, ang_degree):
    x_r, w_r_gl = scipy.special.roots_legendre(n_rad)
    k_max = 30.0
    k_rad = k_max * (x_r + 1.0) / 2.0
    w_krad = w_r_gl * k_max / 2.0
    ang_grid = AngularGrid(degree=ang_degree)
    k_points = np.einsum('i,jk->ijk', k_rad,
                         ang_grid.points).reshape(-1, 3)
    k_rad_expanded = np.repeat(k_rad, len(ang_grid.weights))
    k_weights = (np.outer(w_krad, ang_grid.weights).ravel()
                 * k_rad_expanded ** 2)
    return k_points, k_weights
```

Listing 2: Spherical k-space grid construction.

3.5 NUFFT Implementation Details

All NUFFT evaluations use the type-3 transform from the FINUFFT library, which computes:

$$F_j = \sum_{m=1}^{M} c_m \, e^{-i\,\mathbf{k}_j \cdot \mathbf{x}_m}, \quad j = 1, \ldots, N_k,$$

where both the source points $\{\mathbf{x}_m\}$ and target points $\{\mathbf{k}_j\}$ are non-uniform. The tolerance parameter is set to $\varepsilon = 10^{-8}$ for the one-step method and $\varepsilon = 10^{-5}$ for the two-step method. The tighter tolerance in the one-step case is necessary because the entire ERI depends on a single transform, whereas the two-step method benefits from error cancellation between the forward and inverse transforms.

```python
def solve_nufft_1step(r_p, r_w, dens, k_pts, k_weights,
                      is_spherical=False):
    c = np.ascontiguousarray(dens * r_w, dtype=np.complex128)
    F_k = finufft.nufft3d3(x, y, z, c, kx, ky, kz,
                           isign=-1, eps=1e-8)
    K2 = kx**2 + ky**2 + kz**2
    if is_spherical:
        # k^2 in weights cancels 1/k^2 Coulomb kernel
        base_weights = k_weights / K2
        integral = np.sum(base_weights * 4*np.pi
                          * np.abs(F_k)**2) / (2*np.pi)**3
    else:
        safe_K2 = np.where(K2 < 1e-20, np.inf, K2)
        integral = np.sum(k_weights * (4*np.pi / safe_K2)
                          * np.abs(F_k)**2) / (2*np.pi)**3
    return integral, len(r_p) + len(k_pts)
```

Listing 3: One-step NUFFT ERI computation (spherical k-grid branch).

3.6 Benchmarking Protocol

Each method is tested across the grid parameter values $N \in \{2, 4, 8, 12, 16, 32, 64, 130\}$. For each run, we record:

- **Absolute error**: $|\mathrm{ERI}_{\mathrm{numerical}} - \mathrm{ERI}_{\mathrm{exact}}|$,
- **Wall-clock time**: measured via `time.perf_counter()`,

- **Peak memory**: tracked via Python's `tracemalloc` module.

Direct summation methods are capped at $N \leq 32$ to avoid excessive runtime from $\mathcal{O}(N^6)$ scaling (since the grid has N^3 points for Cartesian).

4 Results

Figure 1 presents the benchmark results across all seven methods as functions of the grid parameter N.

eri_benchmark.png

Figure 1: Benchmark results for seven ERI computation methods. **Left:** Absolute error versus grid parameter N (log scale). **Centre:** Wall-clock computation time (log scale). **Right:** Peak memory usage (log scale). Methods: (1) Direct Cartesian, (2) Direct Becke, (3) FFT Uniform, (4) NUFFT 1-step Cartesian k, (5) NUFFT 1-step Spherical k, (6) Poisson 2-step Cartesian k, (7) Poisson 2-step Spherical k.

4.1 Accuracy Convergence

The left panel of fig. 1 reveals markedly different convergence behaviours:

Direct methods (1–2). The direct Cartesian method (Method 1) converges slowly, with the error plateauing near 10^{-1} even at $N = 32$. This is because the uniform grid poorly resolves the cusp-like density near the nuclei, and the $1/r$ singularity in the Coulomb kernel introduces discretisation artifacts. The direct Becke method (Method 2) converges significantly faster because the atom-centred grid concentrates points where the density is largest. However, both methods share $\mathcal{O}(N^2)$ complexity in the number of grid points, limiting their practical utility.

Uniform FFT (3). The FFT Poisson solver shows moderate convergence, reaching errors around 10^{-1} at large N. Its accuracy is fundamentally limited by the uniform grid: the Nyquist frequency $k_{\max} = \pi/\Delta x$ grows only linearly with N, and the grid cannot adapt to the non-uniform density distribution. The FFT does benefit from excellent computational scaling, however.

NUFFT one-step methods (4–5). Both Cartesian and spherical k-grid variants benefit from the Becke grid in real space, giving them better resolution near the nuclei. The spherical k-grid variant (Method 5, dashed green) converges more smoothly than the Cartesian k-grid variant (Method 4) because the spherical quadrature is naturally suited to the isotropic Coulomb kernel, and the k^2 cancellation avoids singularity issues. At $N = 64$, Method 5 reaches errors near 10^{-2}.

Poisson two-step methods (6–7). The two-step Poisson solver with a spherical k-grid (Method 7, dashed red) achieves the best convergence of all methods, reaching errors below 10^{-3} at moderate grid sizes and continuing to improve. This superiority arises because the two-step approach constructs a smooth real-space potential $V^{(g)}(\mathbf{r})$ before performing the final integration, which is more amenable to accurate numerical quadrature than the oscillatory k-space integrand in the one-step approach. The Cartesian k-grid variant (Method 6) also converges well but is hampered by the singularity at $\mathbf{k} = \mathbf{0}$.

4.2 Computational Cost

The centre panel of fig. 1 shows wall-clock times on a logarithmic scale.

The direct methods exhibit the steepest scaling, with times reaching nearly one second at $N = 32$ due to $\mathcal{O}(N^6)$ operations for the Cartesian grid. The FFT method scales as $\mathcal{O}(N^3 \log N)$ and is the fastest at small N, benefiting from highly optimised FFTW-based implementations. However, it requires N^3 grid points, so the absolute cost grows rapidly.

The NUFFT methods (4–7) occupy an intermediate regime. Their cost is dominated by the type-3 NUFFT, which scales as $\mathcal{O}((M + N_k)\log(M + N_k))$ where M is the number of real-space points and N_k the number of k-space points. The spherical k-grid variants (5, 7) use fewer k-points than their Cartesian counterparts because the Lebedev grid is more compact than a Cartesian cube in k-space, giving them a consistent speed advantage. At large N, all NUFFT methods maintain sub-second runtimes.

4.3 Memory Usage

The right panel shows peak memory consumption. Direct methods are memory-intensive because they construct the full $N \times N$ distance matrix. The FFT method uses moderate memory for the 3D grid and its transform. The NUFFT methods are the most memory-efficient of the spectral approaches, since they avoid storing dense matrices and operate on sparse, non-uniform point sets. Methods using spherical k-grids consistently use less memory than those with Cartesian k-grids because the spherical grid requires far fewer points to achieve comparable angular resolution.

5 Discussion

5.1 Why the Poisson Two-Step Method Excels

The superior convergence of Method 7 (Poisson two-step with spherical k-grid) can be understood from several mathematical perspectives.

First, the two-step formulation separates the forward transform, kernel application, and inverse transform into distinct stages. This means the NUFFT approximation error enters twice but affects different quantities: the forward transform approximates $\widetilde{G}(\mathbf{k})$ and the inverse transform reconstructs $V^{(g)}(\mathbf{r})$. Because $V^{(g)}(\mathbf{r})$ is a smooth Coulomb potential (no cusps or singularities away from the nuclei), the inverse NUFFT achieves high accuracy, and the final real-space integration on the Becke grid is exact to high order.

Second, the spherical k-grid exploits the isotropy of the Coulomb kernel. The $4\pi/k^2$ factor depends only on $|\mathbf{k}|$, so a grid with Gauss–Legendre points in the radial direction achieves exponential convergence for smooth radial integrands. In contrast, a Cartesian k-grid wastes points in the corners of the cube where $|\mathbf{k}|$ is large, and the integrand is negligible.

Third, the analytical cancellation of k^2 between the Jacobian and the Coulomb kernel (sections 1 and 2) means the effective integrand in k-space is $4\pi\,|\widetilde{G}(\mathbf{k})|^2$ weighted by the base quadrature weights, without any singular factor. This makes the quadrature inherently well-conditioned.

5.2 Limitations of the Direct and FFT Methods

The direct methods, while conceptually simple, suffer from two compounding issues. The $\mathcal{O}(N^2)$ pairwise summation is inherently expensive, and the discretisation of the $1/r$ singularity introduces systematic errors that diminish slowly with grid refinement. The self-interaction exclusion (setting the distance matrix diagonal to infinity) is a necessary but imperfect remedy.

The uniform FFT method is limited by its inability to adapt to the molecular geometry. All points of the density profile, including the near-nuclear cusps, must be resolved by the same uniform spacing. This results in either insufficient resolution (if N is small) or excessive memory and compute (if N is large enough to resolve the cusps).

5.3 Cartesian versus Spherical k-Space Grids

A consistent finding across both the one-step and two-step methods is the superiority of spherical k-grids. This can be quantified by comparing the number of grid points required:

a Cartesian k-grid with N_k points per dimension uses N_k^3 total points, whereas the spherical grid uses approximately $N_{\text{rad}} \times N_{\text{ang}}$ points, where N_{ang} is typically a few hundred (from the Lebedev grid). For the same accuracy, the spherical grid requires one to two orders of magnitude fewer points.

Furthermore, the Cartesian k-grid introduces a regularisation problem at $\mathbf{k} = \mathbf{0}$ (the Coulomb singularity), which is handled by setting $K^2 = \infty$ for small values. This ad hoc treatment introduces a systematic bias, whereas the spherical grid avoids the issue entirely.

5.4 Numerical Stability Considerations

The NUFFT tolerance parameter ε plays a critical role. In the one-step method, the entire ERI depends on a single forward NUFFT, so a tight tolerance ($\varepsilon = 10^{-8}$) is required. In the two-step method, the tolerance can be relaxed to $\varepsilon = 10^{-5}$ because errors in the forward and inverse transforms partially cancel when computing the inner product. This tolerance asymmetry is both a practical advantage (faster NUFFT execution) and a theoretical expectation from the variational nature of the energy functional.

6 Conclusion

This study benchmarked seven numerical methods for computing electron repulsion integrals on the H_2 molecule. Our principal findings are as follows:

1. The **Poisson two-step NUFFT with spherical k-space quadrature** (Method 7) achieves the best accuracy–efficiency trade-off, converging below 10^{-3} absolute error with moderate grid sizes and competitive wall-clock times. The analytical cancellation of the k^2 Jacobian with the Coulomb kernel singularity is the key mathematical insight enabling this performance.

2. **Becke molecular grids** in real space consistently outperform uniform Cartesian grids across all methods that use them, confirming that adaptive, atom-centred quadrature is essential for molecular integrals.

3. **Spherical k-grids** outperform Cartesian k-grids in both accuracy and efficiency, due to better alignment with the isotropic Coulomb kernel and natural singularity cancellation.

4. **Direct summation** methods, while useful as baselines, are impractical beyond small grid sizes due to $\mathcal{O}(N^2)$ scaling.

5. The **uniform FFT** serves as a middle ground but is fundamentally limited by its inability to adapt to non-uniform density distributions.

references